Xinting Liao | Curriculum Vitae

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I am a final-year Ph.D. candidate majoring in Computer Science of Zhejiang Univerisity. My research interests include trust-worthy machine learning, privacy-preserving computing, and foundation models.

Contact

Web links:

- Homepage
- Google Scholar
- ORCID
- DBLP
- GitHub
- Semantic Scholar

Education and Professional Service

Zhejiang University — **Hangzhou, China**
Doctor of Science in Computer Science — *Sep. 2020 – May 2025 (Expected)*

- *Supervisor*: **Xiaolin Zheng** & **Chaochao Chen**.

Sichuan University — **Chengdu, China**
Bachelor of Science in Software Engineering — *Sep. 2016 – May 2020*

Experience

NExT++ Lab in National University of Singapore — **Singapore**
Visiting Student — *Jun. 2024 – Present*

- *Supervisor*: **Tat-Seng Chua**.
- *Task*: Federated recommender system, Privacy in foundation models.

Text Intelligence Lab in Westlake University — **Hangzhou, China**
Visiting Student — *Oct. 2019 – May. 2020*

- *Supervisor*: **Yue Zhang**.
- *Task*: Mining concept stocks based on language model (BERT).

Medical Image Lab in Sichuan University — **Chengdu, China**
Undergraduate Intern — *Mar. 2018 – Jun. 2019*

- *Supervisor*: **Yi Zhang**.
- *Task*: System of breast lesion diagnosis.

Awards

Prizes:

- The 3rd Chinese Graduate Artificial Intelligence Innovation Competition (Third Prize) — *2021*
- Longfor Scholarship — *2023*
- Wen-Chixiang Scholarship — *2024*
- Outstanding Graduate Student — *2023, 2024*
- Five Virtues Student — *2024*

Professional Service

Program Committee Member (Reviewer)

- ACM MM,
- ACL ARR,
- EMNLP ARR,
- ICML,
- NeurIPS,
- ICLR,
- AISTATS

Selected publication list

Conference papers:

1. *PPGenCDR: A Stable and Robust Framework for Privacy-Preserving Cross-Domain Recommendation.*
   Xinting Liao, Weiming Liu, Xiaolin Zheng, Binhui Yao, Chaochao Chen.
   In Proc. of **AAAI 2023**
2. *Federated Probabilistic Preference Distribution Modelling with Compactness Co-Clustering for Privacy-Preserving Multi-Domain Recommendation.*
   Weiming Liu, Chaochao Chen, **Xinting Liao**, Mengling Hu, Jianwei Yin, Yanchao Tan, Longfei Zheng.
   In Proc. of **IJCAI 2023**
3. *HyperFed: Hyperbolic Prototypes Exploration with Consistent Aggregation for Non-IID Data in Federated Learning.*
   Xinting Liao, Weiming Liu, Chaochao Chen, Pengyang Zhou, Huabin Zhu, Yanchao Tan, Jun Wang, Yue Qi.
   In Proc. of **IJCAI 2023**
4. *Joint Local Relational Augmentation and Global Nash Equilibrium for Federated Learning with Non-IID Data.*
   Xinting Liao, Chaochao Chen, Weiming Liu, Pengyang Zhou, Huabin Zhu, Shuheng Shen, Weiqiang Wang, Mengling Hu, Yanchao Tan, Xiaolin Zheng.
   In Proc. of **ACM MM 2023**
5. *Robust Representation Learning with Reliable Pseudo-labels Generation via Self-Adaptive Optimal Transport for Short Text Clustering.*
   Xiaolin Zheng, Mengling Hu, Weiming Liu, Chaochao Chen, **Xinting Liao**.
   In Proc. of **ACL 2023**
6. *Learning Accurate and Bidirectional Transformation via Dynamic Embedding Transportation for Cross-Domain Recommendation.*
   Weiming Liu, Chaochao Chen, **Xinting Liao**, Mengling Hu, Yanchao Tan, Fan Wang, Xiaolin Zheng, Yew-Soon Ong.
   In Proc. of **AAAI 2024**
7. *Rethinking the Representation in Federated Unsupervised Learning with Non-IID Data.*
   Xinting Liao, Weiming Liu, Chaochao Chen, Pengyang Zhou, Fengyuan Yu, Huabin Zhu, Binhui Yao, Tao Wang, Xiaolin Zheng, Yanchao Tan.
   In Proc. of **CVPR 2024**
8. *Reducing Item Discrepancy via Differentially Private Robust Embedding Alignment for Privacy-Preserving Cross Domain Recommendation.*
   Weiming Liu, Xiaolin Zheng, Chaochao Chen, Jiahe Xu, **Xinting Liao**, Fan Wang, Yanchao Tan, Yew-Soon Ong.
   In Proc. of **ICML 2024**
9. *User Distribution Mapping Modelling with Collaborative Filtering for Cross Domain Recommendation.*
   Weiming Liu, Chaochao Chen, **Xinting Liao**, Mengling Hu, Jiajie Su, Yanchao Tan, Fan Wang.
   In Proc. of **WWW 2024**
10. *FOOGD: Federated Collaboration for Both Out-of-distribution Generalization and Detection.*
    Xinting Liao, Weiming Liu, Pengyang Zhou, Fengyuan Yu, Jiahe Xu, Jun Wang, Wenjie Wang, Chaochao Chen, Xiaolin Zheng.
    In Proc. of **NeurIPS 2024**
11. *FedGF: Enhancing Structural Knowledge via Graph Factorization for Federated Graph Learning.*
    Pengyang Zhou, Chaochao Chen, Weiming Liu, **Xinting Liao**, Fengyuan Yu, Zhihui Fu, Xingyu Lou, Jun Wang, Wu Wen, Xiaolin Zheng
    In Proc. of **WSDM 2024**

Patents:

1. *Privacy protection cross-domain recommendation method based on generative model.*
   Chaochao Chen, **Xinting Liao**, Xiaolin Zheng.
   Application Number **CN202310742959.1**
2. *A federal modeling method and system based on non-independent identically distributed data.*
   Chaochao Chen, Xiaolin Zheng, **Xinting Liao**.
   Application Number **CN202310757831.2**
3. *The invention relates to a federal learning method of non-independent identically distributed data based on Nash equilibrium aggregation.*
   Chaochao Chen, **Xinting Liao**, Xiaolin Zheng.
   Application Number **CN117313836A**

Xinting Liao | Publication list

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Publications: **17** papers published in top-tier conferences
2 other publications (preprints, etc)
(out of which **5** first-authored papers).

Total number of citations: >196 — **h-index**: 9

Web links to list services: Home Page; Google Scholar; Semantic Scholar; GitHub; DBLP; ORCID.

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1. *PPGenCDR: A Stable and Robust Framework for Privacy-Preserving Cross-Domain Recommendation.*
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In Proc. of **ACM MM 2023**
5. *Differentially Private Sparse Mapping for Privacy-Preserving Cross Domain Recommendation.*
Weiming Liu, Xiaolin Zheng, Chaochao Chen, Mengling Hu, **Xinting Liao**, Fan Wang, Yanchao Tan, Dan Meng, Jun Wang.
In Proc. of **ACM MM 2023**
6. *Joint Internal Multi-Interest Exploration and External Domain Alignment for Cross Domain Sequential Recommendation.*
Weiming Liu, Xiaolin Zheng, Chaochao Chen, Jiajie Su, **Xinting Liao**, Mengling Hu, Yanchao Tan.
In Proc. of **WWW 2023**
7. *Robust Representation Learning with Reliable Pseudo-labels Generation via Self-Adaptive Optimal Transport for Short Text Clustering.*
Xiaolin Zheng, Mengling Hu, Weiming Liu, Chaochao Chen, **Xinting Liao**.
In Proc. of **ACL 2023**
8. *Learning Accurate and Bidirectional Transformation via Dynamic Embedding Transportation for Cross-Domain Recommendation.*
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In Proc. of **AAAI 2024**
9. *Rethinking the Representation in Federated Unsupervised Learning with Non-IID Data.*
Xinting Liao, Weiming Liu, Chaochao Chen, Pengyang Zhou, Fengyuan Yu, Huabin Zhu, Binhui Yao, Tao Wang, Xiaolin Zheng, Yanchao Tan.
In Proc. of **CVPR 2024**
10. *Mitigating Catastrophic Forgetting in Large Language Models with Self-Synthesized Rehearsal.*
Jianheng Huang, Leyang Cui, Ante Wang, Chengyi Yang, **Xinting Liao**, Linfeng Song, Junfeng Yao, Jinsong Su.
In Proc. of **ACL 2024**
11. *Reducing Item Discrepancy via Differentially Private Robust Embedding Alignment for Privacy-Preserving Cross Domain Recommendation.*
Weiming Liu, Xiaolin Zheng, Chaochao Chen, Jiahe Xu, **Xinting Liao**, Fan Wang, Yanchao Tan, Yew-Soon Ong.
In Proc. of **ICML 2024**
12. *CE-RCFR: Robust Counterfactual Regression for Consensus-Enabled Treatment Effect Estimation.*
Fan Wang, Chaochao Chen, Weiming Liu, Tianhao Fan, **Xinting Liao**, Yanchao Tan, Lianyong Qi, Xiaolin Zheng.
In Proc. of **KDD 2024**

13. *User Distribution Mapping Modelling with Collaborative Filtering for Cross Domain Recommendation.*
Weiming Liu, Chaochao Chen, **Xinting Liao**, Mengling Hu, Jiajie Su, Yanchao Tan, Fan Wang.
In Proc. of **WWW 2024**

14. *FOOGD: Federated Collaboration for Both Out-of-distribution Generalization and Detection.*
Xinting Liao, Weiming Liu, Pengyang Zhou, Fengyuan Yu, Jiahe Xu, Jun Wang, Wenjie Wang, Chaochao Chen, Xiaolin Zheng.
In Proc. of **NeurIPS 2024** (Recently Accepted)

15. *Mining User Consistent and Robust Preference for Unified Cross Domain Recommendation.*
Xiaolin Zheng, Weiming Liu, Chaochao Chen, Jiajie Su, **Xinting Liao**, Mengling Hu, Yanchao Tan.
In Proc. of **TKDE 2024**

16. *FedGF: Enhancing Structural Knowledge via Graph Factorization for Federated Graph Learning.*
Pengyang Zhou, Chaochao Chen, Weiming Liu, **Xinting Liao**, Fengyuan Yu, Zhihui Fu, Xingyu Lou, Jun Wang, Wu Wen, Xiaolin Zheng
In Proc. of **WSDM 2024**

17. *Learning Uniform Clusters on Hypersphere for Deep Graph-level Clustering.*
Mengling Hu, Chaochao Chen, Weiming Liu, Xinyi Zhang, **Xinting Liao**, Xiaolin Zheng
Preprint

18. *Federated Learning for Short Text Clustering.*
Mengling Hu, Chaochao Chen, Weiming Liu, **Xinting Liao**, Xiaolin Zheng
Preprint

19. *Enhancing Progressive Diagnosis Prediction in Healthcare with Continuous Normalizing Flows.*
Yanchao Tan, Hengyu Zhang, Zihao Zhou, Guofang Ma, Fan Wang, Weiming Liu, **Xinting Liao**, Vicki S Hertzberg, Carl Yang
Companion Proceedings of WWW 2024

20. *Enhancing Dual-Target Cross-Domain Recommendation with Federated Privacy-Preserving Learning.*
Zhenghong Lin, Wei Huang, Hengyu Zhang, Jiayu Xu, Weiming Liu, **Xinting Liao**, Fan Wang, Shiping Wang, Yanchao Tan
In Proc. of **IJCAI 2024**

21. *Joint Similarity Item Exploration and Overlapped User Guidance for Multi-Modal Cross-Domain Recommendation .*
Weiming Liu, Chaochao Chen, Jiahe Xu, **Xinting Liao**, Fan Wang, Xiaolin Zheng, Zhihui Fu, Ruiguang Pei, Jun Wang
In Proc. of **WWW 2025** (Recently Accepted)

22. *FedGOG: Federated Graph Out-of-Distribution Generalization with Diffusion Data Exploration and Latent Embedding Decorrelation.*
Pengyang Zhou, Chaochao Chen, Weiming Liu, **Xinting Liao**, Wenkai Shen, Jiahe Xu, Zhihui Fu, Jun Wang, Wen Wu David, Xiaolin Zheng
In Proc. of **AAAI 2025** (Recently Accepted)

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